

Eviction Policy Part 2

Prepared By: Dr. D. P. Singh

Least Frequently Used (LFU) Overview

- **LFU (Least Frequently Used) eviction**

is a cache eviction policy that **removes** the items that are accessed the **least often** when the cache runs out of memory.

- Each cached item maintains an access frequency counter.

Working:

Key	Access Count
A	100
B	25
C	3
D	1

If memory is full and a new item needs to be inserted, LFU evicts the item with the lowest access count (in this example, **D**).

LFU in Redis

- Redis supports LFU through these eviction policies:
- **allkeys-lfu**
 - Evict the least frequently used key from all keys.
- **volatile-lfu**
 - Evict the least frequently used key only **among keys that have an expiration (TTL)**.

Example (Redis)

Suppose the Redis has:

user:1 accessed 500 times

user:2 accessed 100 times

user:3 accessed 2 times

- When memory reaches **maxmemory** and a new key is inserted, Redis is more likely to evict user:3.

Implementation Details (Redis)

- Redis does **not** store an exact access count forever because that would be memory-intensive.
- Instead, it uses:
 - A logarithmic probabilistic counter
 - Counter aging/decay over time
- This prevents an item that was popular months ago from staying in cache forever.
- For example: Yesterday: key A accessed 10,000 times
 Today: key A never accessed
- Its LFU score gradually decays so newer hot keys can replace it.

Note: Memcached primarily used **LRU (Least Recently Used)** eviction rather than LFU.

- Thank You